

Chapter 1

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May 08, 2024

1.6. Cardinality

Problem 1. Show $\mathbb{Q}$ is countably infinite

Proof. It suffices to show there is a injective map from $\mathbb{Q} \rightarrow \mathbb{N} \times \mathbb{N}$ since $\mathbb{N} \times \mathbb{N}$ is countable and $\mathbb{Q}$ is infinite. Define $f : \mathbb{Q} \rightarrow \mathbb{N} \times \mathbb{N}$ where $\forall \frac{p}{q} \in \mathbb{Q} \ni p, q \in \mathbb{Z} \ni q \neq 0$ where wlog $\gcd(p, q) = 1$.

$$f\left(\frac{p}{q}\right) := \begin{cases} (2p, 2q) & \text{if } p, q \geq 0 \\ (2p, -2q + 1) & \text{if } p \geq 0 \text{ and } q < 0 \\ (-2p + 1, 2q) & \text{if } q \geq 0 \text{ and } p < 0 \\ (-2p + 1, -2q + 1) & \text{if } p, q < 0 \end{cases}$$

$$\forall \frac{p}{q}, \frac{r}{s} \in \mathbb{Q} \ni f\left(\frac{p}{q}\right) = f\left(\frac{r}{s}\right).$$

1. If $p, q \geq 0 \implies r, s \geq 0$, thus $2p = 2r \implies p = r$, and $2q = 2s \implies q = s$.
2. Cases 2, 3, and 4 are the same.

f is injective. Thus $\mathbb{Q}$ is countably infinite. □

Problem 2. Show that the maps f and g of Examples 1 and 2 are bijections.

Claim 2a. f is a bijection.

$$f(n) := \begin{cases} 2n & \text{if } n > 0 \\ -2n + 1 & \text{if } n \leq 0 \end{cases}$$

Proof. $\forall n, m \in \mathbb{Z} \ni f(n) = f(m)$, if $n > 0 \implies 2n = f(m)$. Assume the contrary that $m \leq 0$, then $2n = -2m + 1 \implies m = n - \frac{1}{2} \implies m \notin \mathbb{Z}$ which is a contradiction. Thus $m > 0$. Thus $2n = 2m \implies m = n$. If $n \leq 0 \implies -2n + 1 = f(m)$. Assume the contrary that $m > 0 \implies 2n - 1 = 2m \implies m = n - \frac{1}{2} \implies m \notin \mathbb{Z}$ which is a contradiction. Thus $2n - 1 = 2m - 1 \implies n = m$. Thus f is an injective function.

$$\forall y \in \mathbb{Z}_+.$$

1. If y is even: Let $x = \frac{y}{2}$. $f(x) = y$.
2. If y is odd: Let $x = -\frac{y-1}{2}$, $f(x) = -2x + 1 = y$.

Thus f is surjective. □

Problem 3. Let X be the two-element set $\{0, 1\}$. Show there is a bijective correspondence between the set $2^{\mathbb{Z}_+}$ and the cartesian product X^ω .

Proof. Let $f : 2^{\mathbb{Z}_+} \rightarrow X^\omega$ where $A \in 2^{\mathbb{Z}_+}$ gets sent to a element of X^ω where the i th component is 1 iff $i \in A$ otherwise the i th component is 0.

$\forall A, B \in 2^{\mathbb{Z}_+}, f(A) = f(B)$. $\forall i \in A$, then the i th component of $f(A)$ is 1. Thus the i th component of $f(B)$ is 1. Thus $i \in B$. Thus $A \subseteq B$. $\forall j \in B$, then j th component of $f(B)$ is 1. Thus the j th component of $f(A)$ is 1. Thus $j \in A$. Thus $B \subseteq A$. $A = B$. Thus f is injective.

$\forall (a_1, \dots, a_i, \dots) \in X^\omega$. Let $A = \{x \in \mathbb{Z}_+ \mid a_x = 1\}$. $f(A) = (a_1, \dots, a_i, \dots)$. Thus f is surjective. Thus f is a bijection. □

Infinite Sets and the Axiom of Choice

Problem 9.1. Define an injective map $f : \mathbb{Z}_+ \rightarrow X^\omega$, where X is the two-element set $\{0, 1\}$ without using the choice axiom.

Proof. Let $f : \mathbb{Z}_+ \rightarrow X^\omega$ where $m \mapsto (\delta(1 - m), \dots, \delta(i - m), \dots)$.

Suppose $\forall m, n \in \mathbb{Z}_+ \ni f(m) = f(n)$ that means $f(m)$ and $f(n)$ agree on the m and n th entry. That means there are 1s on the m and n th entry. For $f(m)$, there is a 1 only on the m th entry similarly for $f(n)$ will only have a 1 on the n th entry. Thus there is a 1 on the m th entry and n th entry which means $m = n$. □

Problem 9.2. Find if possible a choice function for each of the following collections, without using the choice axiom:

1. The collection A of nonempty subsets of $\mathbb{Z}_+$
2. The collection B of nonempty subsets of $\mathbb{Z}$
3. The collection C of nonempty subsets of the rational $\mathbb{Q}$
4. The collection D of nonempty subsets of X^ω where $X = \{0, 1\}$

Proof. 1. Let $c : A \rightarrow \bigcup_{B \in A} B \ni B \rightarrow \min(B)$.

2. Let f be the bijection from $\mathbb{Z} \rightarrow \mathbb{Z}_+$ seen in problem 1.6.1. Let $c : B \rightarrow \bigcup_{A \in B} A \ni A \rightarrow \min(f(A))$

For the rest do the same. First map to $\mathbb{Z}_+$ and then take the min which works since $\mathbb{Z}_+$ is well ordered. □

Problem 9.3. Suppose that A is a set and $\{f_n\}_{n \in \mathbb{Z}_+}$ is a given indexed family of injective functions

$$f_n := [1, \dots, n] \rightarrow A$$

Show A is infinite. Can you define a injective function from $f : \mathbb{Z}_+ \rightarrow A$ without using the choice axiom.

Proof. Assume the contrary A is finite. Let $m := |A|$. There is a bijective map f_m from $[1, \dots, m] \rightarrow A$. Thus since f_{m+1} is injective. There is a injective map from $[1, \dots, m+1] \rightarrow [1, \dots, m]$, which is a contradiction. Thus A is infinite.

Define f recursively. Let $a_0 = f_1(1)$ and let

$$\rho(h : [1, \dots, m] \rightarrow A) = f_m(\min k \in \mathbb{Z}_+ | k \leq n+1, f_m(k) \notin h([1, \dots, m-1]))$$

Since ρ is well defined f_n is injective. Thus f is defined in the recursive function and thus injective. $\square$

Well Order Sets